HOCHSCHILD & CYCLIC (CO)HOMOLOGY SS22

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This lecture course aims to study Hochschild (co)homology and cyclic (co)homology of associative algebras along with the perspectives of some applications in Lie algebras, non-commutative geometry, loop space topology and symplectic geometry. We will cover the following topics.

1. Basic Category Theory

- Grothendieck universe, definition of category and basic concepts.
- Functors and natural transformations.
- Representable functors and Yoneda lemma.
- Adjoint functors.
- Limits and colimits.

2. Homological Algebra

- Additive and Abelian categories.
- Category of complexes, projective and injective objects.
- Triangulated categories.
- Localization of categories.
- Derived categories and derived functors.

3. Hochschild (Co)Homology

- Hochschild complexes, bar resolutions, Hochschild (co)homology.
- Cup product, Gerstenhaber bracket, cross product, Künneth formula, Hodge decomposition.
- Morita invariance.
- Calabi-Yau algebras, Poincaré duality, Batalin-Vilkovisky structure.

4. Cyclic (Co)Homology

- Tsygan bicomplex, Connes mixed complex, cyclic complex, cyclic (co)homology.
- Connes exact sequence, Morita invariance.
- Hochschild-Kostant-Rosenberg theorem and a glimpse into non-commutative geometry.

5. Lie Algebra (Co)Homology

- Lie algebra (co)homology and Chevalley-Eilenberg complex.
- Hamiltonian actions and moment maps.
- Whitehead lemmas.
- Loday-Quillen-Tsygan theorem.

6. Hochschild Invariant of Loop Space

- Hochschild (co)homology of dg algebras.
- Loop spaces, Burghelea-Fiedorowicz-Goodwillie theorem and Jones theorem.
- Outlook towards Hocschild invariant in string topology and symplectic topology.